

Did you identify Figure 1.10 as a rat (or a mouse)? If you did, you were influenced by the clearly ratlike or mouselike figure you observed initially. But people who first observe Figure 1.13 on page 11 usually identify Figure 1.10 as a man. (Try this demonstration on some other people.) The **rat-man demonstration** shows that the subject's expectations about the meaning of a stimulus can influence their perception of the stimulus. We used the phenomenological method to show this because the data we needed was the subject's description of what he or she perceived.

Another example of an experiment in which subjects are asked to describe what they see is one by Steven Palmer (1975), which used the stimuli in Figure 1.18. Palmer first presented a context scene such as the one on the left and then briefly flashed one of the target pictures on the right. When Palmer asked subjects to identify the object in the target picture, they correctly identified an object like the loaf of bread (which is appropriate to the kitchen scene) 80 percent of the time, but correctly identified the mailbox or the drum (two objects that don't fit into the scene) only 40 percent of the time. As in the rat-man demonstration, this experiment also uses the phenomenological method to show how perception can be influenced by a person's expectations. We will see, as we study the relationship between stimuli in the world and perception, that asking people to describe what they perceive is an important tool for studying the perceptual process.

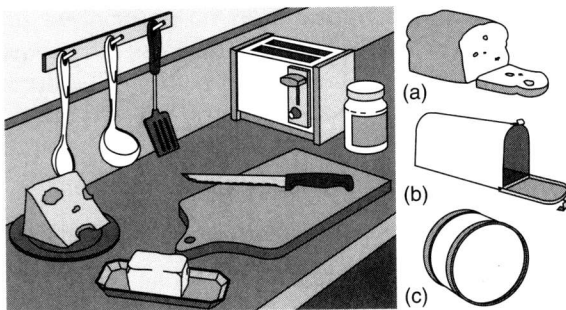


Figure 1.8
Stimuli used in Palmer's (1975) experiment. The scene at the left is presented first, and the observer is then asked to identify one of the objects on the right.

Although results from the phenomenological method can be valuable, they are usually qualitative in nature—they provide description but not precise measurement. Often, however, it is useful to be able to establish a quantitative relationship between the stimulus and perception. To achieve this, we look to the psychophysical methods. We begin by describing a group of methods called the “Classical” psychophysical methods because they were developed in the 19th century.

The Classical Psychophysical Methods: Measuring Thresholds

In the 19th century, the first scientific psychologists developed a number of methods called the **classical psychophysical methods** to measure thresholds—the smallest amount of stimulus energy necessary for an observer to detect a stimulus. Gustav Fechner (1801–1887) was a pioneer in developing the classical psychophysical methods. He proposed that it would be possible to demonstrate a relationship between the functioning of the body and experience by having people report their experiences as physical stimuli are systematically changed. Fechner cleverly picked an experience that was relatively easy to define—whether or not a person can detect a stimulus. In 1860, Fechner published his book *Elements of Psychophysics*, in which he focused on measuring a quantity called the absolute threshold.

The Absolute Threshold The **absolute threshold** is the smallest amount of stimulus energy necessary for an observer to detect a stimulus. Fechner described the following three methods for measuring the absolute threshold: the *method of limits*, the *method of adjustment*, and the *method of constant stimuli*.

To measure the threshold by using the **method of limits**, the experimenter presents stimuli in either ascending or descending order, as shown in Figure 1.9, which indicates the results of an experiment that measures a person's threshold for seeing a light.

On the first series of trials, the experimenter presents a light with an intensity of 105, and the observer

	1	2	3	4	5	6	7	8
Intensity	↓	↑	↓	↑	↓	↑	↓	↑
105	Y						Y	
104	Y		Y		Y		Y	
103	Y		Y		Y		Y	
102	Y		Y		Y		Y	
101	Y		Y		Y		Y	Y
100	Y	Y	Y	Y	Y		Y	Y
99	Y	N	Y	N	Y	Y	Y	Y
98	N	N	Y	N	N	N	N	Y
97		N	N	N		N		N
96		N		N		N		N
95		N		N		N		N
Crossover values	98.5	99.5	97.5	99.5	98.5	98.5	98.5	97.5

Threshold = Mean of crossovers = 98.5

Figure 1.9

The results of an experiment to determine the threshold using the method of limits. There are eight series of trials in this experiment, the descending trials alternating with the ascending trials. In the first series, the observer indicates by answering “no” at an intensity of 98 that he no longer sees the light. This change from “yes” at 99 to “no” at 98 is the crossover point, and the threshold value for this run is taken as the mean between 99 and 98, or 98.5. The procedure is then repeated in reverse, starting below the threshold and increasing the intensity until the subject says “yes.” Both the descending and the ascending presentations are repeated a number of times, and the threshold is calculated as the mean of the crossover values for each run. The threshold is therefore 98.5 in this experiment.

indicates by a “yes” response that he sees the light. This response is indicated by a Y at an intensity of 105 on the table. The experimenter then decreases the intensity, and the observer makes a judgment at each intensity until he responds “no,” that he did not

see the light. This change from “yes” to “no” is the *crossover point* and the threshold for this series is taken as the mean between 99 and 98 or 98.5. By repeating this procedure a number of times, starting above the threshold half the time and starting below the threshold half the time, the threshold can be determined by calculating the average of all of the crossover points.

In the **method of adjustment**, the stimulus is slowly changed as either the observer or the experimenter adjusts the stimulus intensity in a continuous manner (as opposed to the stepwise presentation for the method of limits) until the observer can just barely detect the stimulus. This just barely detectable intensity is then taken as the absolute threshold. This procedure can be repeated several times and the threshold determined by taking the average setting. Having the observer adjust the stimuli has the advantage of giving the observer an active role in the process, thus maximizing the probability that he or she will stay awake during the entire experiment!

In the **method of constant stimuli**, five to nine stimuli are typically presented by the experimenter in random order. The results of a hypothetical determination of the threshold for seeing a light are shown in Figure 1.11. The data points in this graph were determined by presenting six light intensities 10 times each and determining the percentage of times that



Figure 1.10

Did you see a “rat” or a “man”? Looking at the more ratlike picture in Figure 1.7 increased the chances that you would see this one as a rat. But if you had first seen the man version (see Figure 1.13), you would have been more likely to perceive this figure as a man. (From Bugelski & Alampay, 1961.)

the subject perceives each intensity. The results indicate that the light with an intensity of 150 is never detected, the light with an intensity of 200 is always detected, and lights with intensities in between are sometimes detected and sometimes not detected. The threshold is usually taken as the intensity that results in detection on half the trials, so in this case the threshold is an intensity of 180.

The choice between the methods of limits, adjustment, and constant stimuli is usually determined by the accuracy that is needed and the amount of time available. The method of constant stimuli is the most accurate method but takes the longest time, whereas the method of adjustment is the least accurate but the fastest.

When Fechner published *Elements of Psychophysics*, he not only described his methods for measuring the absolute threshold, but also described the work of Ernst Weber (1795–1878), a physiologist, who, a few years before the publication of Fechner's book, measured another type of threshold, the *difference threshold*.

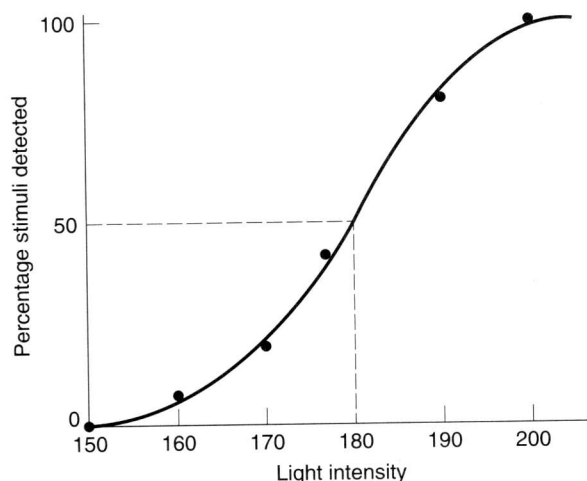


Figure 1.11

Results of a hypothetical experiment in which the threshold for seeing a light is measured by the method of constant stimuli. The threshold, the intensity at which the light is seen on half of its presentations (as indicated by the dashed line), is 180 in this experiment.

The Difference Threshold The difference threshold, which is also called the **just noticeable difference (JND)**, is the smallest difference between two stimuli that a person can detect. To measure the difference threshold, Weber had subjects lift a small “standard” weight and then lift a slightly heavier “comparison” weight and judge which was heavier (Figure 1.12). When the difference between the standard and comparison weights was small, subjects found it difficult to detect the difference in the weights, but they easily detected larger differences. That much is not surprising, but Weber went further. He found that the size of the JND depended on the size of the standard weight. For example, the JND for a 100-gram weight is 5 grams (a subject could tell the difference between a 100- and 105-gram weight, but smaller differences remained undetected), and the JND for a 200-gram weight is 10 grams. Thus, as the magnitude of the stimulus increases, so does the size of the JND.



D E M O N S T R A T I O N

Measuring the JND

By doing a simple experiment with two boxes of wooden matches, you can demonstrate to yourself that the JND gets larger as the standard stimulus gets larger. Have a friend place 10 matches in one box (the standard) and 11 in the other (the comparison). Comparing the weights of the two boxes with your eyes closed, try to decide which box is heavier. After making this judgment, repeat this procedure; if you can't correctly judge which is heavier on three out of three trials, have your friend place another match in the comparison box, and try again. Continue this procedure until you can consistently judge which box is heavier. If, for example, you can consistently tell that the comparison box is heavier when it contains 12 matches, then the JND equals 2 matches. Now repeat the above procedures, but start with 20 matches in the standard box and 21 in the comparison. Since the JND gets larger as the weight of the standard gets larger, you should find that the JND for the 20-match standard is larger than the JND for the 10-match standard. ●

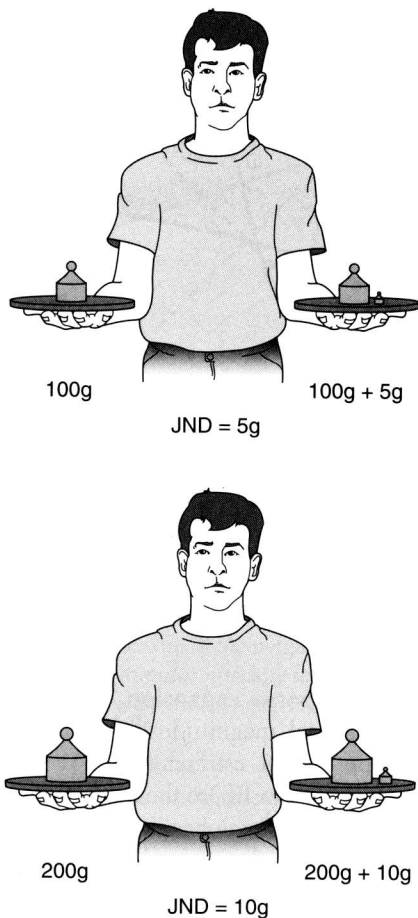


Figure 1.12

The just noticeable difference (JND). The person can detect the difference between a 100-gram weight and a 105-gram comparison weight but cannot detect a smaller difference, so the JND is 5 grams. With a 200-gram weight, the comparison weight must be 210 grams before the person can detect the difference, so the JND is 10 grams. Note that for both the 100- and 200-gram weights, the ratio of JND to weight is the same. This ratio is the Weber fraction.

Research on a number of senses has shown that the JND is larger for larger standard stimuli, and that, over a fairly large range of intensities, the ratio of the JND to the standard stimulus is constant. This relationship, which is based on Weber's research, was stated mathematically by Fechner as $JND/S = K$ and

was called **Weber's law**. K is a constant called the *Weber fraction*, and S is the value of the standard stimulus. Applying this equation to our example of lifted weights, we find that, for the 100-gram standard, $K = 5/100 = 0.05$, and that, for the 200-gram standard, $K = 10/200 = 0.05$. Thus, in this example, Weber's fraction (K) is constant. In fact, numerous modern investigators have tested Weber's law and found that it is true for most senses, as long as the stimulus intensity is not too close to the threshold (Engen, 1972; Gescheider, 1976).

Fechner's proposal of three psychophysical methods for measuring the absolute threshold and his statement of Weber's law for the difference threshold were extremely important events in the history of scientific psychology because they demonstrated something that many people thought was impossible: the quantification of mental activity. This was a notable achievement at the time, and the methods developed in the 1800s are still used today to measure absolute and difference thresholds. In addition to being used to determine thresholds in research laboratories, simplified versions of the classical psychophysical methods have been used to measure people's detail vision when determining the prescription for glasses, or people's hearing when testing for possible hearing loss.

The classical psychophysical methods were developed to measure absolute and difference thresholds. But what about perceptions that occur above threshold? Most of our everyday experience consists of perceptions that are far above threshold, when we can easily see and hear what is happening around us.

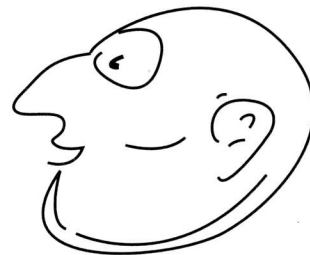


Figure 1.13

Man version of the rat-man stimulus. (Adapted from Bugelski & Alampay, 1961.)

In order to measure these above-threshold perceptions, S.S. Stevens developed a technique called *magnitude estimation*.

Magnitude Estimation: Measuring Magnitude above Threshold

If we double the intensity of a light, does it look twice as bright? If we double the intensity of a tone, does it sound twice as loud? Although a number of researchers, including Fechner, proposed equations that related perceived magnitude and stimulus intensity, it wasn't until 1957 that S. S. Stevens developed a technique called *scaling* or **magnitude estimation** that accurately measured this relationship (Stevens, 1957, 1961, 1962).

Magnitude estimation is relatively simple: The experimenter first presents a "standard" stimulus to the observer (let's say a light of moderate intensity) and assigns it a value of, say, 10; he or she then presents lights of different intensities, and the observer is asked to assign a number to each of these lights that is proportional to the brightness of the light. If the light appears twice as bright as the standard, it gets a rating of 20; half as bright, a 5; and so on. Thus, each light intensity has a brightness assigned to it by the observer.

The results of a magnitude estimation experiment on brightness are plotted in Figure 1.14. This graph plots the means, for a number of observers, of the magnitude estimates of the brightness of a light versus the intensity of the light. You can see from the way this curve bends down that doubling the intensity does not double the perceived brightness. Doubling the intensity causes only a small change in perceived brightness, particularly at higher intensities. This result is called **response compression**. As intensity is increased, the responses increase, but not as rapidly as the intensity. To double the brightness, it is necessary to multiply the intensity by about 9.

Figure 1.14 also shows the results of magnitude estimation experiments for the length of a line, and for the sensation caused by an electric shock presented to the finger. The electric shock curve bends up, indicating that doubling the strength of a shock more than doubles the sensation of being shocked.

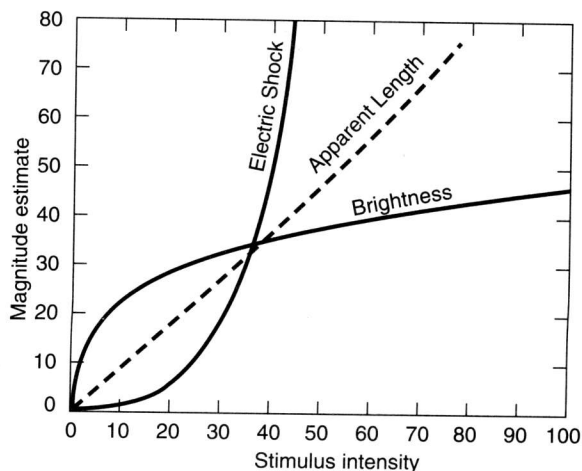


Figure 1.14

Curves showing the relationship between perceived magnitude and stimulus intensity for electric shock, line length, and brightness. (Adapted from Stevens, 1962.)

This is called **response expansion**. As intensity is increased, perceptual magnitude increases more than intensity. The length estimation curve is straight, with a slope of close to 1.0, so the magnitude of the response almost exactly matches increases in the stimulus (i.e., doubling the line length doubles the observer's estimate of the length of the line).

The beauty of the relationships derived from magnitude estimation is that the relationship between the intensity of a stimulus and our perception of its magnitude follows the same general equation for each sense. We show this by plotting the *logarithm* of the magnitude estimates versus the *logarithm* of the stimulus intensity, which causes all three curves to become straight lines (Figure 1.15). These functions are called **power functions** and are described by the equation $P = KS^n$. Perceived magnitude, P , equals a constant, K , times the stimulus intensity, S , raised to a power, n . This relationship is called **Stevens' power law**.

The power n , the exponent of the power law, indicates the slope of the lines in Figure 1.15. Remembering our discussion of the three types of curves in Figure 1.14, we can see that the curve that shows response compression has a slope of less than 1.0, the

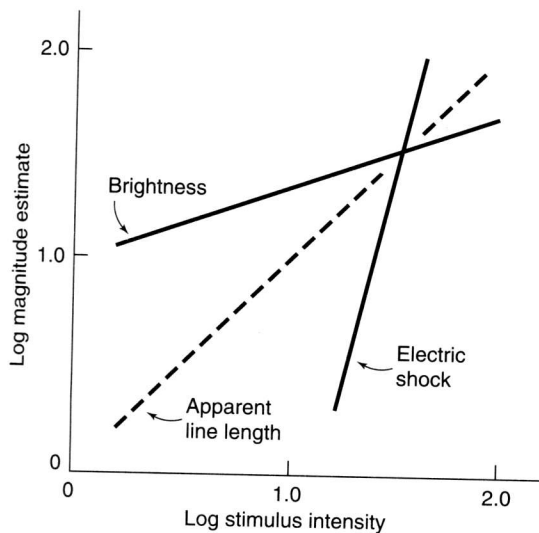


Figure 1.15

The three curves from Figure 1.14 plotted on log-log coordinates. Taking the logarithm of the magnitude estimates and the logarithm of the stimulus intensity turns the curves into straight lines. (Adapted from Stevens, 1962.)

straight line has a slope of about 1.0, and the curve that shows response expansion has a slope of greater than 1.0. Thus, the relationship between response magnitude and stimulus intensity is described by a power law for all senses, and the exponent of the power law indicates whether doubling the stimulus intensity causes more or less than a doubling of the response.

These exponents not only illustrate that all senses follow the same basic relationship, they also illustrate how the operation of each sense is adapted to how organisms function in their environment. Consider, for example, our experience of brightness. The light intensities present outside on a bright sunny day could be as high as 1,000 to 1 or greater if we compare, say, the sunlight reflected from the hood of a just polished car to the dark pavement underneath the car. If our perception of brightness spanned as large a range as the light intensities, we might be faced with such high contrasts and bright glare that looking at the scene would be difficult. However, the exponent for brightness is about 0.6, so changing intensity by a factor of

about 9 or 10 just doubles the brightness, keeping our experience of contrasts within manageable limits.

The opposite situation occurs for electric shock, which has an exponent of 3.5 so that just small increases in shock intensity cause large increases in pain. This rapid increase in pain even to small increases in shock intensity serves to warn us of impending danger, and we therefore tend to withdraw even from weak shocks.

Using the techniques we have just described to determine relationships between stimuli and perception is one way to study perception. Another way is to use the physiological approach.

THE PHYSIOLOGICAL APPROACH: LINKING STIMULATION AND NEURAL FIRING

What are the physiological mechanisms of perception? Modern research designed to answer this question has focused on determining the relationship between the stimulus and electrical signals called *nerve impulses*, and on determining the relationship between these nerve impulses and perception. Before describing the nature of these nerve impulses, let's look at some of the early history that led to this approach.

The Physiological Approach: Early History

Our modern ideas about the physiological basis of perception are descended from a long line of speculation and research regarding the physiological workings of the mind. Early research and speculations in this area focused on determining the anatomical structures involved in the operation of the mind. In the 4th century BC the philosopher Aristotle (384–322 BC) stated that